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The Role of Materials Science and Engineering on Developing More Efficient and Innovative Medical Devices with Faster and Developed Treatment Processes and Reduced Material Waste

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Abstract:

This paper will focus on the benefits of materials science and engineering to medicine and healthcare. It will explore the ways materials science processes such as materials characterization, pro-

cessing, selection and new materials development contributes to the development of smart medical devices. The paper will also address the significant materials like nanomaterials and biomaterials to create medical innovations

like artificial organs, chemical sensors, biosensors, and devices utilized in the pharmaceutical industry. This paper will argue that materials science and engineering is beneficial both in the current healthcare system and in the future medical applications since it enables quicker production, decreases the amount of wasted material, and increases the efficiency of the product which enables better diagnosis and treatment for patients. It will also discuss the advantages of materials science including improved accuracy, endurance, and decreased costs, while discussing the drawbacks of the developed devices. Furthermore, this paper will highlight the future applications of the development of medical devices such as point-of-care testing and remote treatment. Even though material science and engineering will play a vital role in medicine in the future, research done by scholars show that the material science and engineering field should become more common to improve and provide global equitable medical diagnosis and treatment.

Keywords: materials science and engineering, medicine, materials research, materials processing, artificial organs, sensors, pharmaceuticals.

1-Introduction:

Have you ever wondered whether or not nanoparticles could detect a tumor in the human body? Do you ever consider that it is possible to get information about a person's health by sim-

ply analyzing their sweat? These ideas, and many more are all possible with materials science and engineering know-how. In the 21st century, especially after industrialization, the importance of materials has increased substantially. The intention and the aim of the first industrial revolution started during the mid-1800s, was to mine and drill materials to push forth mass production. The companies continued to produce more to increase their income, but they did not consider the world or the environment specifically. In the fourth industrial revolution, or today in Industry 4.0, people have noticed that they cannot continue to consume raw and natural resources in such high amounts because of the concerns about climate change. The carbon emissions and footprints are significant problems to the future of our world which primarily result from burning fossil fuels. Fortunately, with the developments in technology, these problems can be addressed by materials science. The purpose of materials science is to research and utilize materials to achieve sustainability. Therefore, material scientists want to think about non-dependence on the mining of raw materials, and it wants to create an opportunity to evaluate the potential of waste materials and all kinds of "garbage" which makes up a large proportion of all materials we throw away into landfills.

Therefore, it is important to develop biocompatible materials for the human body in the venture to reduce the amount of wasted materials. Since

materials information leads engineers in designing smart devices, nowadays there are many innovative devices being developed.

For example, Professor Nathan Gianneschi, Northwestern Univ., Department of Chemistry, utilizes nanoparticles and nanomaterials to detect tumors in the human body [19]. He designed nanoparticles that can construct therapeutic particles to detect tumor [19].

Moreover, analyzing sweat and detecting diseases may have seemed surprising, but this is among other innovation, which become available by the developments in materials science. An eyeglass shaped gadget monitors glucose amount in sweat using glucose-sensitive pads [2]. It is found that blood-sweat glucose levels and sweat concentration are highly correlated in

healthy individuals [2].

These two examples clearly suggest the wide application of materials science and engineering in medicine. Materials Science and Engineering is focused on all natural and synthetic materials, including their synthesis, processing, classification, and improvement for technological usages [14]. Materials scientists are employed in every part of technology, from the research of materials to classification and development to assist creating medical innovations [14].

In addition to the involvement of materials science in diverse aspects of human life, it plays a significant role in one of the biggest issues people are facing: waste management.

According to the World Bank, each year, 2.01 billion tons of solid are wast-

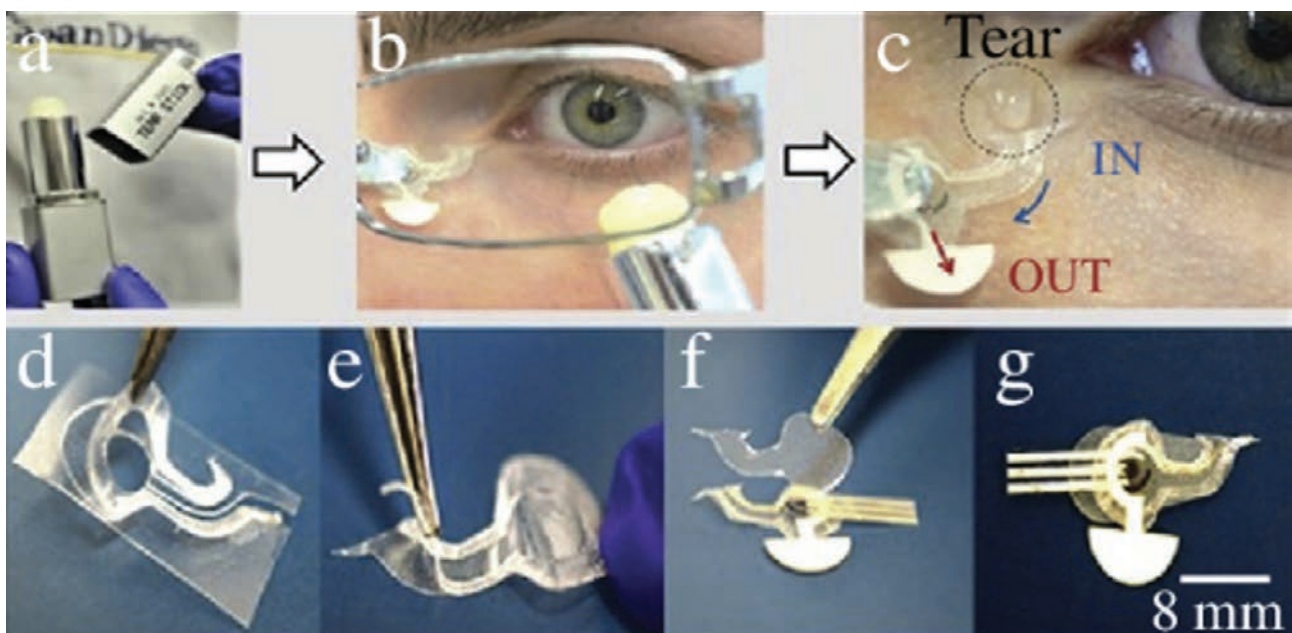


Image 1.
Eyeglass Shaped Sweat Biosensor [2]

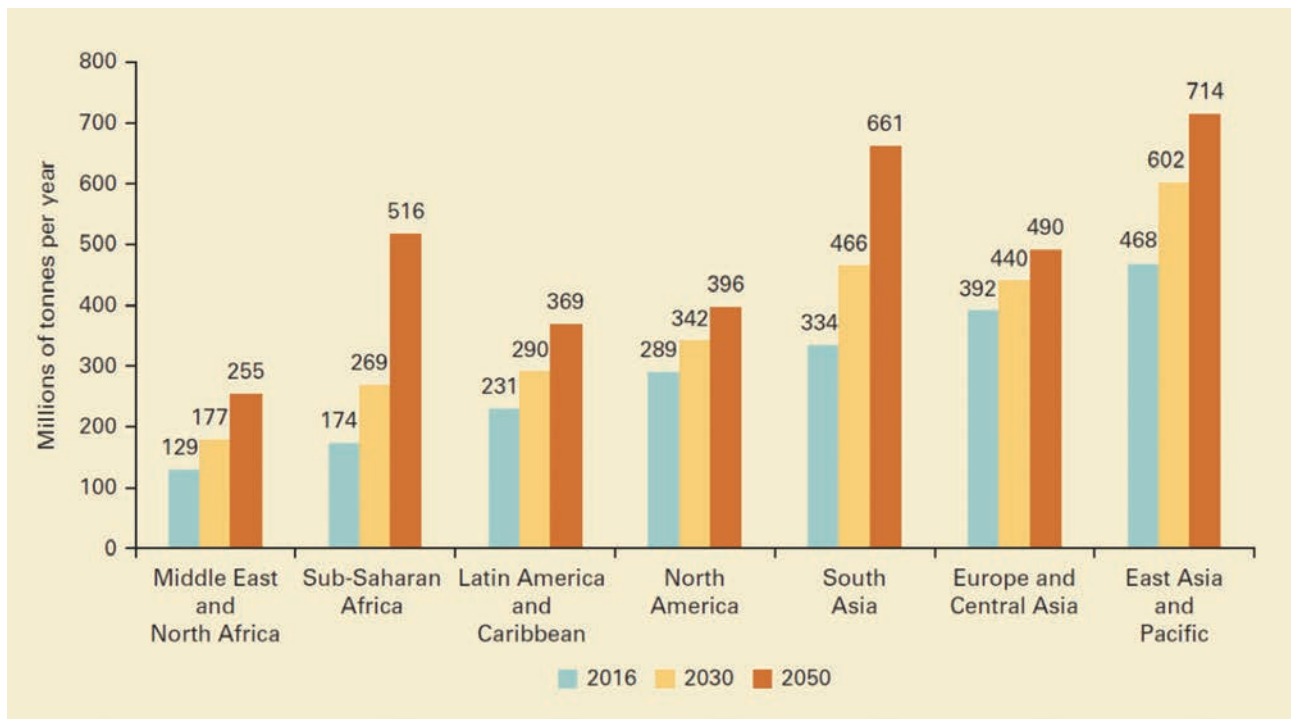


Image 2.
Predicted waste amount by 2050 [25]

ed, and by 2050 it will grow to 3.40 billion tons [25]. In fact, this vast amount of waste is even more than double the population growth over the same period which proves the severity of the problem. Nevertheless, this problem can be addressed by materials science since it enables reusing materials.

By noticing the severity of this problem, nowadays, people are seeking to find sustainable and environmentally friendly ways to produce new products. Therefore, materials science enables people, researchers and companies to search various materials to find the procedure with the lowest impact on the environment. Research and the development of new materials can lead to a smaller environmental footprint and sustainable technologies that meet

United Nations (UN) Sustainable Development Goals (SDGs).

Furthermore, by applying knowledge about materials, new products with higher performance and efficiency are produced which increase the living standards of the citizens. The increased knowledge about materials also leads to innovations which improve society's welfare. In the 21st century, it is impossible to innovate without the use of artificial intelligence which also plays a significant role in materials science.

Artificial intelligence over the last decade has become an extremely important technological tool that has enabled the humans to achieve much more software-based applications that range in many sectors from the business sector to medicine. While Alan Tur-

ing perhaps did not envision the boom in the practice of what he had questioned during the 1950s, it is now clear that artificial intelligence has made its way into every aspect of our lives. AI and its subsets such as machine learning and deep learning along with many others such as neural networks and expert systems can assist humans in processing immense amounts of data. AI can help achieve various tasks because it is able to classify and make predictions about our data and assist many experts in various industries to help consumers better. By removing additional barriers like high cost, time, and effort, machine learning (ML) which is a specialized and well-liked subset of AI, now being applied in materials research, offers an alternative for direct testing and reduces the complexity of finding the attributes of materials.

As technological innovations increase, researchers start to utilize new ways in materials science and engineering. For the first time, researchers utilized a quantum computer to provide precise results from materials science simulations and simplify the data acquired [10]. With technological improvements, the materials science and engineering field will acquire more accurate and complex information obtained from classical digital computers which will be utilized to enhance the treatment processes [10].

In addition to artificial intelligence, materials science and engineering improve welfare and the healthcare system by creating precise solutions enabled by nanomaterials and bioma-

terials. Materials scientists can acquire special combinations of the desired attributes by investigating the chemical compositions' design, processing methods, and materials functions [20]. Therefore, the materials science and engineering field foster future breakthroughs like artificial organs, tissue engineering, organ-on-a-chip technology, sensors, wearable technologies, remote healthcare systems, developed imaging tools, and smart devices useful in the pharmaceutical industry. By playing a significant role in developing such medical innovations, materials science and engineering is beneficial for disease prevention, diagnosis, and treatment since it allows for improving the performance of the existing materials while accelerating the treatment processes and reducing the amount of wasted materials. Therefore, there should be more emphasis on this field and the new generation should be encouraged accordingly when they plan their career.

2.0 The Impact of Nanomaterials in Medicine

Can you imagine a 1-meter ruler that is made smaller to one-billionth? That is extremely small, but nowadays a popular field called nanotechnology enables people to produce nanoscale materials. "Nanotechnology is the understanding and control of matter at the nanoscale, at dimensions between approximately 1 and 100 nanometers, where unique phenomena enable novel applications" [17]. At the nanoscale, matter obtains different physical, bi-

ological and chemical features which makes this field appropriate for materials science and engineering.

Nanotechnology makes it possible for materials scientists and engineers to change the structures of materials like heat and electricity conductivity, durability, reactivity and magnetism [17]. Thus, it extends materials science and engineering by providing materials with various beneficial features. After understanding the property changes, material scientists study to find new applications for these materials.

Even though nanotechnology is utilized in many fields, it can create enormous benefits and improvements in medicine through developed medical tools, knowledge and therapies. Nanomedicine is the application of nanotechnology in medicine to produce more precise solutions for diagnosis and treatment [17]. Materials science and engineering is important in nanomedicine since it provides discovery, characterization, production and modelling of nanoscale materials which are called nanomaterials. Some types of nanomaterials such as fullerene, carbon nanotubes, and nanocrystals are all beneficial in medicine from drug delivery to diagnosis [17].

As an example of their wide applications in medicine, fullerenes can be given. Fullerene is an appropriate material for diagnosis and therapy because of its electronic properties. Therefore, various research is done to implement this beneficial material in medicine innovatively. Much research is done to utilize fullerenes to develop MRI con-

trast agents. Furthermore, as fullerene can stabilize immune effector cells to prevent the proinflammatory mediators, it can be used in the treatment of numerous diseases, including arthritis, asthma, and sclerosis [6]. Lastly, the most important research leads to the development of a new class of fullerene-based theragnostic, allowing diagnostic capabilities to detect and execute cancer cells [6].

Another nanomaterial used in medicine is graphene which is an allotrope of carbon used to design neural interface devices. These implanted, electrode-based devices connect to the nervous system and record brain activity [22]. This device is especially utilized to develop motor symptoms of patients who suffer from Parkinson's disease [22]. Instead of alloys, materials science and engineering enables graphene to be used because of its 2D structure, transparency, MRI compatibility, and flexibility [22]. Since materials scientists search for the characteristics of materials, they can offer the most suitable materials to be used. Therefore, the amount of waste is decrease because graphene replaced less effective materials which cannot endure for a long time.

Moreover, with the assistance of the knowledge acquired from materials science and engineering, today several medical tools which are better at diagnosis and imaging are produced. These tools are utilized in personalized treatment and give more successful therapy rates. For example, the engineering of nanomaterials leads to improvements in

gene sequencing technologies [17]. This innovation will decrease the costs and increase the speed of molecule detection which will be very beneficial for the patients who need urgent treatment. It will also decrease the amount of sample required, so less will be wasted.

Nanotechnology is also a novel approach to biomedical imaging. Specific characteristics of nanoscale materials like magnetic and chemical properties allow creating imaging probes with multiple functions, including multi-modal imaging across various scanning equipment, greater contrast enhancement, richer spatial information, and controlled biodistribution [27]. These advancements have provided benefits like the ability to locate tumors, precise disease progression assessment, and tailored medication [27].

Topically applied surface-enhanced resonance Raman ratiometric spectroscopy (TAS3RS) is an example of nanoscale diagnostics [27]. Folate receptors which are the membrane-bound surface proteins are utilized in this imaging method to find tumor lesions as small as 370 micrometers [27]. TAS3RS is dependent on photons scattering inelastically, which produces extremely high sensitivity. Therefore, it permits high-accuracy early diagnosis, making it a viable tool for fighting cancer [27]. This method can be applied locally, which has the benefit of avoiding the bloodstream and any potential toxicity issues with circulating nanoprobe [27].

Nanotechnology and nanomaterials have a high potential in developing

medicine by bringing advances in the treatment of diseases. Since bacteria and viruses cause many diseases and they are nanosized, using nanotechnology will play a very central role in the future too.

Even in the 20th century, famous physicist Richard Feynman noticed the importance of nanotechnology in medicine and offered to develop nanorobotic surgical assistance to heal heart disease [9]. He stated “[i]t would be interesting in surgery if you swallow the surgeon. You put the mechanical surgeon inside the blood vessel, and it goes into the heart and looks around.” [9]. His idea refers to how nanorobots (Image 3) can be placed inside the body to help detect and treat medical conditions. When using nanorobots to cure heart disease, it will spot the valve with the fault and function to treat. These nanorobots can also be applied to cure other dysfunctional organs.

When inserted into the body through the circulatory system or cavities, a surgical nanorobot controlled or guided by a human surgeon could function as a semiautonomous on-site surgeon inside the human body [15]. Nano computers would lead the nanorobots, execute plans, and interact with other computers [15]. Even though there are not any nanorobots that can be used extensively, researchers and engineers are working to develop this practical and innovative technology, which may be common in hospitals within 10 years.

As proved by the given examples, nanotechnology and nanomaterials

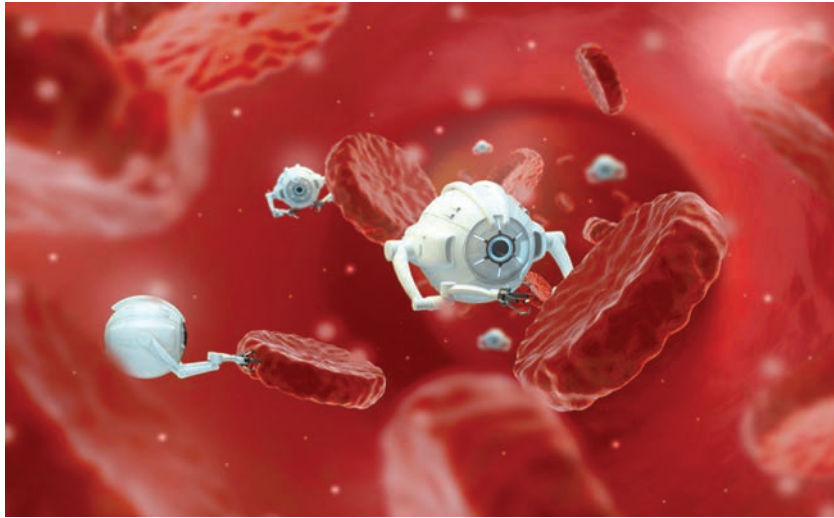


Image 3.
Doctors Inside the Body: Nanorobots [16]

have enormous medical implications from diagnosis to treatment of the disease by enhancing production of innovative devices with reduced waste materials. Nevertheless, nanotechnology is not the only opportunity for scientists to develop efficient medical devices. In addition to nanomaterials, the developed technology provides people to utilize biomaterials to improve their welfare through improving diagnosis and treatment processes.

2.1 Utilizing Biomaterials to Develop more Effective Devices with Higher Biocompatibility with the Human Body

“Biomaterial is a substance that has been engineered to interact with biological systems, including implanted medical devices, prosthetics, skin-based wearable devices, and drug release materials for diagnosis and therapy” [4]. Because they enable a chance to design novel devices, bio-

materials are significant for raising the quality of life. Therefore, these materials are produced by chemical engineers, biomedical engineers, material scientists and engineers to develop human life. The cutting-edge tools, such as biosensors, implants, and artificial organs, can become more compatible through utilizing biomaterials. However, the appropriate materials should be chosen, and the right manufacturing procedure should be followed for creating these biocompatible gadgets. Biocompatibility is a very important element to consider because unfavorable outcomes and device failure will occur if the material is not biocompatible with the body.

Materials scientists and engineers want to develop optimized, biocompatible, well-structured, and efficient devices, so they should combine the natural systems into synthetic constructs. Materials scientists and engineers are inspired by natural processes, so com-

binning natural processes into new technologies is a possible by biomaterials. Nowadays, biomaterials are used in medicine extensively from replacing damaged parts of the body to healing injuries. Biomaterials are used to develop implants, human tissues, nanoparticles, biosensors and drug delivery systems. [3]. Because of materials' different chemical and physical properties, each material will react differently when it is introduced into the body. According to Professor Saltzman, Yale University, Department of Biomedical Engineering and Chemical Environmental Engineering, even though the amount of the response can change in different material designs, the response cannot be eliminated [28]. He also states that the modest reactions demonstrate biocompatibility because few cells are attracted, less protein is absorbed, and thin cellular tissue forms instead of a scar [28]. Even though technology plays an active role in developing new technologies with higher biocompatibility, the same technology puts humanity at a huge risk by damaging the environment. However, several steps and strategies can be taken to utilize technology in an eco-friendly way. Biomaterials enable scientists to develop disposable devices which do not harm the environment. With advanced technology and improved data collection, biodegradable biomaterials are utilized to form disposable medical devices which replaces the materials which ends up as garbage.

As an example of disposable medical tools, dissolvable zinc stents devel-

oped by the National Institute of Biomedical Imaging and Bioengineering (NIBIB) can be given. In the past, metal stents were used to open the vessels. However, these metal stents cause many negative side effects like re-narrowing of the vessel and blood clots [3]. Nevertheless, researchers supported by the NIBIB are creating a bio-absorbable zinc stent that gradually dissolves away without injury, and reduces the typical chronic hazards connected with permanent stents [3]. Also, these disposable stents are more environmentally friendly which is also important to address the waste management problem.

As more research is done, it becomes clear that biomaterials will play an important role in developing medical devices that improve healthcare systems. One day, it may be possible to cure bone deformities, cancer, and heart attacks using synthetic and natural injectable biomaterials which are used for the delivery of medicine, genetic materials and proteins [3].

3.0 Life-Saving Smart Devices: Artificial Organs

As mentioned in the previous section, biomaterials are very important to develop new medical tools. However, biocompatibility is one of the most important factors that scientists must consider. Even though the implanted material should have specialized mechanical and physical functions, the most important criteria when replacing or repairing a body part is biocompatibility with the body. Since the ma-

terial should be compatible with the implanted body, it is a very specific and personalized process. Throughout history, many people have dysfunctional organs which cannot process as a healthy organ. Therefore, people always waited for an organ transplant and sometimes died waiting for a healthy organ. Considering the improved technology, people nowadays can ask if there are any new developments in organ transplantation. It can be also asked how the patients can continue their lives without a functioning organ while waiting for organ transplantation. In fact, the artificial organs enable people to continue their lives.

In the 21st century, technology continues to develop more rapidly than ever before. The availability of new materials, especially nanomaterials and biomaterials, leads to novel medical innovations like artificial organs and implants. When developing artificial organs and implants, biocompatibility should be considered of the highest importance. A human-made device or tissue known as an artificial organ is implanted or integrated into a person to replace the function of the natural organ so the patient can resume their normal life [1]. Nevertheless, providing all of the functions of the natural organ, such as the liver, is very difficult since it requires many liver cells which will make the synthetic organ huge [28]. For this reason, artificial organs are used for assisting the natural organs to function some of their properties [28]. Therefore, the artificial organs enable quicker treatment processes for the

patients since they do not need to wait in suffer for an organ transplant.

Even though organ transplantation is the most effective way of treatment, the lack of donors increases cost, and immunological rejection poses challenges. These challenges lead the scientists to direct their focus to develop artificial organs. Materials science and engineering assist the development of artificial organs by helping with the design and synthesis of materials utilized in artificial organs. Moreover, nanotechnology enables the development of devices for many areas like bone and neural tissue engineering, a field that aims to repair injured tissues and organs [5]. Developing more complicated tissues will lead to creating artificial organs. Furthermore, nanotechnology can be combined with biomaterials to give more effective results in tissue engineering and artificial organs. The dialysis machine invented by Kolff in the 1940s was the first synthetic organ replacement for the kidney [8]. Dialysis is used when the kidneys are injured, yet it is unable to carry out the intricate tasks that the kidneys do [8]. It is a simultaneous diffusion and filtration process with less complexity, but it is quite effective [8]. David Humes, University of Michigan, Department of Division of Nephrology is creating a bio-artificial kidney. The tool integrates a classic haemo-filtration cartridge with a bioreactor chamber that contains a renal tubule assist device [8]. This device delivers 109 beneficial elements which regenerate the good nutrients and send them into the blood [8].

Furthermore, Sean Wu Laboratory from Stanford University works on heart diseases, the biggest cause of death in the United States. They try to understand the mechanism of the heart by using laboratory mice. To build unique replacement tissues to replace the injured heart muscle, they have merged stem cell biology with tissue engineering to create realistic operating heart tissue. [24]. They also utilize the Bio-AssemblyBot (BAB) 3D bioprinter, the first 6-axis robotic arm in the world for quick and sophisticated tissue biofabrication for faster treatment [24].

Even though many developed technologies like BioAssemblyBot (BAB) are utilized in Sean Wu Laboratory, the researchers in this laboratory experiment using mice which is unethical. As discussed by the author of this research paper in another research paper titled "The Impact of Organ on a Chip Technologies on the Advancements of New Therapeutic Approaches", heart on a chip technology can prevent researchers from utilizing the hearts of mice. Heart-on-a-chip technology is a part of the organ-on-chip-technology, and this technology is developed for several organs. "Organ on chips are engineered microchips that replicate the operations of living human organs." [11]. Organ on-a-chip models are very beneficial in the laboratory to replace animal testing. Foundation director of the Wyss Institute for Biologically Inspired Engineering at Harvard, Don Ingber highlights that organ-on-a-chip technology is very useful to replace animal testing [11]. Furthermore, senior staff

scientist Geraldine Hamilton from Harvard University states that organ-on-a-chip models can reflect the conditions for humans more accurately and prevent many resources from being wasted aside from ethical issues [11]. As an alternative to what has been done in Sean Wu Laboratory, professor Mark Saltzman from Yale University also tells his students about Jarvik 7 heart, an artificial heart, in the biomedical engineers and artificial organs course. Jarvik 7 heart can be utilized when the patient's heart is failed [28]. Therefore, it can replace a heart and save the patient's life. Nevertheless, Professor Saltzman highlights that powering the heart is the main problem since it requires high energy [28]. Even though it is not possible for now to make a small battery system that can fit in the patient's body, the current technology addresses this problem by connecting the heart to the batteries with an outside battery system [28]. There are many other challenges that researchers should address while developing artificial organs. The largest problem was finding biocompatible materials for the body. Since biocompatibility is one of the most important factors in treatment, more research should be done in universities to address this problem more quickly for the patients seeking urgent treatment.

By considering the benefits of artificial organ technology and tissue engineering, it seems that the focus will increase on these fields. While some devices are created like dialysis, many artificial organ ideas are in the research

process. However, as technology improves, humanity will encounter many novel artificial organs which enable better treatment with reduced harm to bodies in hospitals and clinics. Since artificial organs create a potential for better treatment, the importance of materials science and engineering increases since this field enables the research and production of biocompatible materials for the human body. The developed artificial organs are very important because they have a potential to save the patient's life by functioning like the natural organs.

3.1 The Effect of Chemical Sensors and Biosensors to Provide Faster Treatment

The developments of artificial intelligence and technology brought many technological devices into everyday life including sensors. Chemical sensors and biosensors are included in many medical devices that improve the diagnosis and treatment. "A biosensor is an analytical device incorporating a biological sensitive recognition element immobilized on a physicochemical transducer to measure analytes." [13]. On the other hand, chemical sensors are equipment that converts chemical data into an analytically usable signal [23]. Since both chemical and biosensors are effective in providing medical information, the usage of these devices has increased substantially. With biosensors, various biological fluids, including blood, sweat, saliva, tears, interstitial fluid, breath, and human volatiles, may now be identified [2].

Nevertheless, it should be noted that collecting the blood by using these sensors, poses a high risk of infection to the patients.

Noticing these challenges, researchers and materials scientists are developing new sensors that can treat patients in a better way. For example, nowadays scientists are working on a digital wound dressing that can deliver oxygen and blood-vessel-promoting biochemical factors [3]. This device will be very beneficial to those who suffer from diabetic ulcers, a disease which decreases the quality of life and leads death [3]. The dressing has wound sensors and actuators in close contact with the skin and use fibers as materials [2]. By utilizing the biomaterials in this technological device, it is expected that this device will be effective in healing. Wearable sensors like the wound biosensor enable monitoring the patient continuously, so wearable sensors may make it possible to switch to personal management at home, which is beneficial for reducing expenses and the crowd in the hospital [2]. Wearable wound biosensors are only an example, which constitutes the huge application areas of materials science, engineering and sensors. Sensors assist people in every aspect you can imagine in medicine.

Sensors have many applications in the human body such as skin gas sensor, breath sensor, tooth sensor, pacifier sensor and many more. As an example to various sensors, a special type of biosensor is being developed by Howard E. Katz, Johns Hopkins University, Department of Materials Science and

Engineering to identify brain damage during heart surgery by responding to glial fibrillary acidic protein (GFAP) which causes brain injuries [12]. According to recent studies, 40 per cent of people have brain injury after going into heart surgery [12]. Therefore, this newly developed sensor is very important since it responds to this protein simultaneously with the surgery by analyzing the patient's blood and emitting a signal when the protein is detected [12]. This sensor can also be utilized outside the surgery as a point-of-care device to treat the patients more rapidly because this sensor allows instantaneous detection and keeps medical professionals from waiting for brain-associated symptoms to occur [12]. The sensors used in medicine are very beneficial since they can replace expensive and complicat-

ed devices, enabling high performance and mobility. Also, the sensors can accommodate a wide range of analytes, so they can be used widely in medicine. Moreover, to maximize the benefits of chips like mobility, degradability, unparalleled accuracies, and simultaneous analysis of several analytes in a single device, the sensors can be constructed on a microfluidic basis [13].

The developed artificial intelligence can be utilized to develop the sensors even more. By adding the wireless systems (Image 4), the patients can be treated out of the hospitals and health-care centers since the usage of artificial intelligence, and wireless systems can store the patient data. This will lead to faster treatment with better results.

Even though this technology has many advantages, there are some

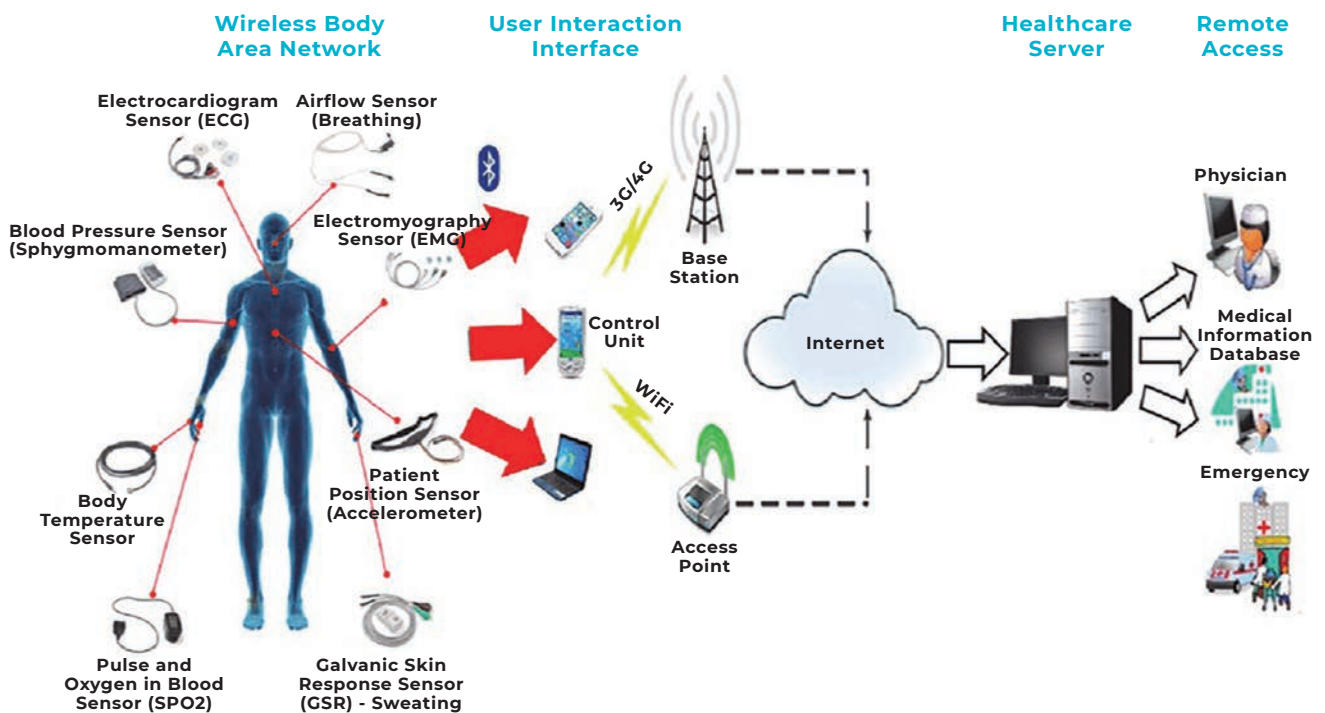


Image 4.
Wearable Sensors [21]

drawbacks which can be addressed by materials science and engineering. The nanomaterials can cause problems related to biocompatibility and toxicity [2]. Moreover, the sensors should be robust enough to resist severe operating conditions [2]. Therefore, the role of materials scientists and engineers is very important to address these challenges, expand the use of sensors in medicine and treat more patients with better methods. These significant challenges can be addressed more effectively if materials science and engineering becomes more common and accessible for the next generation who will shape the future healthcare systems.

4.0 The Benefit of Materials Science in the Pharmaceutical Industry

In addition to the benefits of materials science and engineering to develop medical sensors, this field also plays a vital role in the pharmaceutical industry which focuses on developing drugs for humans. Since materials science and engineering enable material characterization, processing, research, selection, and development of new materials, this field can assist in improving better drugs by using the materials' knowledge acquired. Dynamic light scattering and photon correlation spectroscopy are the methods for identifying materials which are useful to research newly synthesized materials' biocompatibility for effective drug administration [26]. Moreover, the knowledge acquired from the material characterization process is used to design materials with specific attributes to in-

crease efficiency. The data acquired can also give material scientists information about biocompatibility, processing requirements, cost and reliability, which affects material selection. Therefore, by knowing exactly which materials to utilize, the material scientists and engineers directly use that specific material without doing many experiments to test whether the material is suitable to use. This creates many advantages like shortening the production time which enables quicker treatment and prevents waste of materials, which decreases the amount of garbage.

Frank Gu from the University of Toronto is working on developing a nanoparticle which will include a specific nanomedicine placed into the eye to prevent taking eye drops several times during a day [7]. This nanoparticle will adhere to the mucous membrane and release the drug, which decreases the number of drugs used [7]. In addition to nanomaterials, biomaterials are used to solve problems like biological barriers, which cause drugs to be taken out from the body before they can effectively function [18]. Therefore, Mitchell Laboratory from the University of Pennsylvania currently works to address biological barriers in drug delivery which is possible to solve by using biomaterials [18].

As the two studies provided above prove, the material knowledge specifies what kind of material to utilize in specific conditions, in this case, nanomaterial or biomaterial. In fact, the materials science and engineering field provides knowledge about materials or

develops new materials and selects the most compatible materials to develop medical devices. These systems will exactly form the basis of future diagnosis and treatment processes. Even though many functions of materials are unknown, with the developed technology, material scientists will search more profoundly and will find new characteristics that will contribute to designing novel devices which are more compatible with both the human body and the environment. However, in order to discover new characteristics of the materials, more importance should be given to this field both in the universities and in the working environment. In the pharmaceutical industry, materials science prevents long trial and error processes, so fewer materials are wasted, the appropriate devices and drugs are developed quicker, and the patients get better treatment with improved tools and more efficient drugs.

5.0 Conclusion

This research paper emphasized the huge potential of materials science and engineering to develop medicine and form the basis of the future health-care system since they enable more developed and faster treatment and also helps to reduce the amount of waste. In the 21st century, improved technology and artificial intelligence enable the production of many new medical devices. With Industry 4.0, the companies compete to develop more sophisticated devices to provide better treatment systems to society. The paper highlighted the importance of materials science

and engineering in this competition because this field enables knowledge about the materials utilized in smart devices. Furthermore, this research paper discussed more innovative and effective materials like nanomaterials and biomaterials rather than discussing old-fashioned alloys. Especially in medicine, it is important to design materials with specific characteristics, and since nanomaterials can show different characteristics, they provide new application areas in medicine. Also, working on biomaterials is important to create biocompatible tools with the human body to treat the patients without causing any other harm. The paper illustrated the way material science can contribute to medicine by focusing on medical innovations such as artificial organs, chemical and biosensors and smart devices used in drug delivery. By gaining more knowledge about the materials, innovative and effective artificial organs can be produced with the assistance of tissue engineering. Artificial organs are very beneficial while the patient is waiting to receive an organ transplant. Moreover, the research paper discussed the benefits of materials science to develop sensors which can analyze many analytes quickly and simultaneously which shortens the treatment process. The faster treatment is very important for patients who suffer. In addition, the research paper focused on smart devices developed for use in drug delivery which decreases the number of drugs used and prevents waste. Therefore, materials science and engineering can address the environ-

mental problems too. Furthermore, this research paper presented the advantages of materials science and engineering such as efficiency, longevity, decreased environmental footprint, recycling of strategic materials, reduction of material and drug waste, decreased costs, and rapid production. Moreover, materials science and engineering will enable many future applications like point-of-care treatment and remote treatment by utilizing artificial intelligence. All of these advantages and possible future applications contribute to better diagnosis, imaging and treatment which improves the welfare of patients and enhances healthcare systems. However, in order to solve the disadvantages more quickly, materials science and engineering should be more common, and there should be more research done about the characteristics of many unknown materials. It is clear that materials science and engineering will play an active role in shaping both the current and future healthcare systems. Overall, in this research paper, the essential role and potential of materials science and engineering is emphasized to enhance more effective smart devices which enables faster treatment and reduced waste.

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**THE ROLE OF MATERIALS SCIENCE AND ENGINEERING ON
DEVELOPING MORE EFFICIENT AND INNOVATIVE MEDICAL DEVICES
WITH FASTER AND DEVELOPED TREATMENT PROCESSES AND REDUCED
MATERIAL WASTE**

Alp Baran ERKUL*

Abstract

This paper will focus on the benefits of materials science and engineering to medicine and healthcare. It will explore the ways materials science processes such as materials characterization, processing, selection and new materials development contributes to the development of smart medical devices. The paper will also address the significant materials like nanomaterials and biomaterials to create medical innovations like artificial organs, chemical sensors, biosensors, and devices utilized in the pharmaceutical industry. This paper will argue that materials science and engineering is beneficial both in the current healthcare system and in the future medical applications since it enables quicker production, decreases the amount of wasted material, and increases the efficiency of the product which enables better diagnosis and treatment for patients. It will also discuss the advantages of materials science including improved accuracy, endurance, and decreased costs, while discussing the drawbacks of the developed devices. Furthermore, this paper will highlight the future applications of the development of medical devices such as point-of-care testing and remote treatment. Even though material science and engineering will play a vital role in medicine in the future, research done by scholars show that the material science and engineering field should become more common to improve and provide global equitable medical diagnosis and treatment.



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